

RELATIVE MOTION AND FRAMES OF REFERENCE

JEE Advanced Preparation — Physics

Topic 04 | Relative Velocity, River-Boat, Rain Problems, Inertial and Non-Inertial Frames

Concept	Formula	Application
Relative velocity	$v_{AB} = v_A - v_B$	Frame change, min distance
River — shortest time	$t = d/v_b$ (aim perp)	Drift = $v_r * d/v_b$
River — zero drift	$\sin(\theta) = v_r/v_b$ (upstream)	Requires $v_b > v_r$
River — $v_b < v_r$	Min drift angle: $\sin = v_b/v_r$	Cannot cross straight
Rain problem	$v_{rain_wrt_man} = v_{rain} - v_{man}$	Tilt umbrella toward v_{man}
Pseudo force	$F_p = -m*a_{frame}$	Non-inertial frame analysis
Restitution	$e = v_{sep}/v_{app}$	$0 \leq e \leq 1$

Relative motion is the foundation of kinematics in JEE Advanced. Understanding how observers in different frames see the same physical event is essential for solving river-boat problems, rain problems, minimum distance problems, and non-inertial frame analyses.

KEY FORMULAS REFERENCE

Relative position: $r_{AB} = r_A - r_B$ (position of A as seen from B's location)

Relative velocity: $v_{AB} = v_A - v_B$ (velocity of A in frame of B)

Relative acceleration: $a_{AB} = a_A - a_B$ (only if both frames are inertial)

River shortest time: $t = d/v_b$; drift = $v_r * (d/v_b)$; aim directly across river

River zero drift: $\sin(\theta_{upstream}) = v_r/v_b$; actual speed = $\sqrt{v_b^2 - v_r^2}$

River $v_b < v_r$: min drift angle from bank: $\sin(\phi) = v_b/v_r$; cannot cross straight

Rain: $v_{rain_man} = v_{rain_ground} - v_{man}$; umbrella direction = v_{rain_man} direction

Pseudo force: $F_p = -m*a_{frame}$ (add this to each body in accelerating frame)

Apparent weight in lift: $W' = m*(g + a)$ for upward accel a ; $W' = m*(g-a)$ for downward

Min distance: $d_{min} = |r_{AB}| * \sin(\alpha)$; where α = angle between r_{AB} and v_{AB}

SECTION 1: RELATIVE POSITION, VELOCITY, ACCELERATION

Definition: $r_{AB} = r_A - r_B$ means 'position of A as observed from B'.

Differentiating: $v_{AB} = v_A - v_B$ (velocity of A as seen by observer at B).

Differentiating again: $a_{AB} = a_A - a_B$ (valid only when both frames are inertial).

Key properties:

$v_{AB} = -v_{BA}$ (relative velocity is antisymmetric)

$v_{AB} + v_{BC} = v_{AC}$ (relative velocity is transitive / adds like vectors)

For inertial frames only: $a_{AB} = a_A - a_B$

Minimum distance between two moving particles:

Step 1: Treat B as stationary (work in B's reference frame).

Step 2: A moves with velocity $v_{AB} = v_A - v_B$ in this frame.

Step 3: Initial relative position: $r_{AB}(0) = r_A(0) - r_B(0)$.

Step 4: Minimum distance = perpendicular distance from B to the line of A's relative motion.

$d_{\min} = |r_{AB}(0)| \sin(\alpha)$, where α = angle between $r_{AB}(0)$ and v_{AB} .

Step 5: Time to closest approach: $t = |r_{AB}(0)| \cos(\alpha) / |v_{AB}|$

(only valid if $\cos(\alpha) > 0$, i.e., particles are initially approaching).

Vector subtraction graphically:

To find $v_{AB} = v_A - v_B$: draw v_A , then draw $-v_B$ (reverse of v_B) from the same point.

The resultant is v_{AB} . This is the 'velocity triangle' method used in river-boat and rain problems.

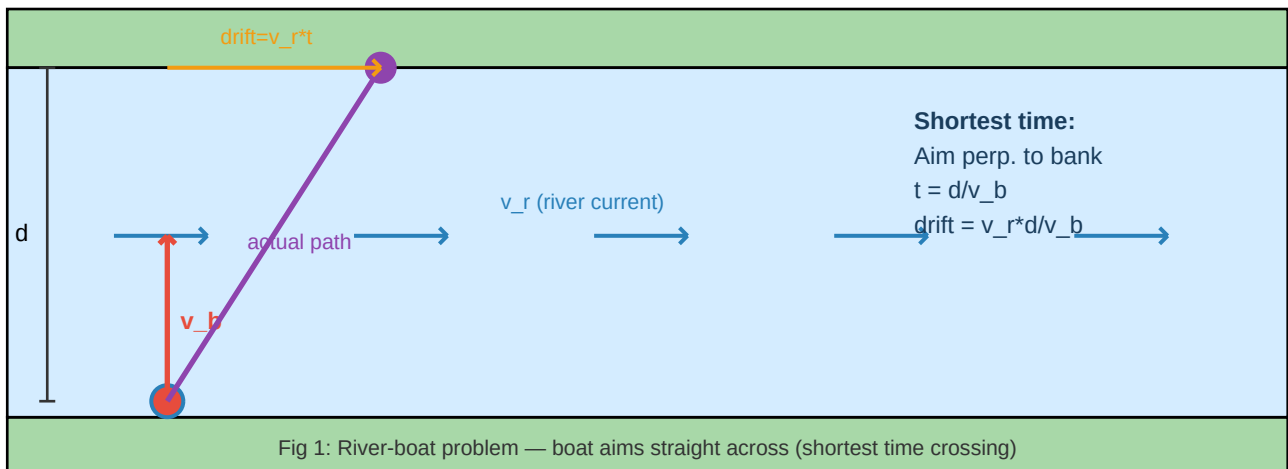
IMPORTANT: $v_{AB} = v_A - v_B$ does NOT mean the speed of A minus the speed of B.

It is a VECTOR subtraction. You must subtract the velocity vectors, not their magnitudes.

Example: $v_A = 3$ m/s east, $v_B = 4$ m/s north. $v_{AB} = 3$ east - 4 north = (3,-4) m/s.

Magnitude of $v_{AB} = 5$ m/s, NOT $3-4 = -1$ m/s.

SECTION 2: RIVER-BOAT PROBLEM — COMPLETE ANALYSIS



Setup: River width = d . River flows right at speed v_r . Boat speed in still water = v_b .
 Actual velocity of boat = v_b (relative to water) + v_r (river velocity).
 $v_{\text{boat_ground}} = v_{\text{b_vector}} + v_{\text{river_vector}}$ (vector sum).

Case 1: Minimum time crossing

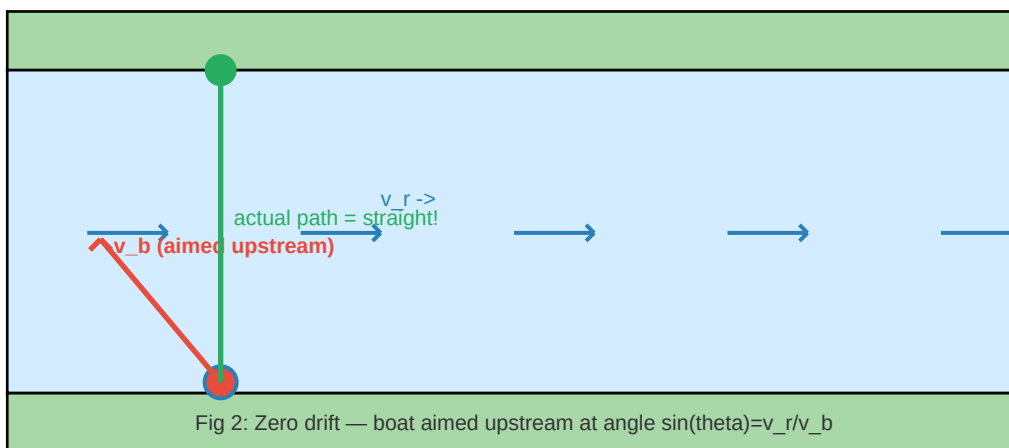
Aim boat directly across (perpendicular to banks). Then:
 Component across river = v_b (full boat speed goes across).
 Component along river = v_r (drift due to current).
 Time: $t = d / v_b$ (minimum possible time).
 Drift: $x = v_r * t = v_r * d / v_b$.
 Actual speed over ground: $v_{\text{ground}} = \sqrt{v_b^2 + v_r^2}$.
 Path angle from perpendicular: $\tan(\phi) = v_r / v_b$.

Case 2: Zero drift (straight-across path) — requires $v_b > v_r$

Aim boat upstream at angle θ where $\sin(\theta) = v_r / v_b$.
 The river component exactly cancels the boat's upstream component.
 Effective across-river speed: $v_{\text{across}} = \sqrt{v_b^2 - v_r^2}$.
 Time: $t = d / \sqrt{v_b^2 - v_r^2}$ (longer than minimum time case).
 Drift: 0 (by definition of this case).

Case 3: $v_b < v_r$ (cannot achieve zero drift)

For minimum drift: aim at angle ϕ to the bank where $\sin(\phi) = v_b / v_r$.
 Minimum drift = $d * \sqrt{v_r^2 - v_b^2} / v_b$.
 The boat always moves downstream in this case; question is minimizing how much.



Comparison of Cases 1 and 2:

Case 1 (min time): $t_1 = d/v_b$; $x_1 = v_r d/v_b$ (drift is non-zero).

Case 2 (zero drift): $t_2 = d/\sqrt{v_b^2 - v_r^2} > t_1$ (takes longer).

Ratio: $t_2/t_1 = v_b / \sqrt{v_b^2 - v_r^2} = 1/\sqrt{1 - (v_r/v_b)^2} \geq 1$.

The zero-drift path always takes more time than the direct crossing.

For minimum distance path (not zero drift, not min time):

This is the path that minimizes the distance traveled over ground.

Angle: $\theta = \arctan(v_r/\sqrt{v_b^2 - v_r^2})$ from upstream (this is the zero-drift angle).

No, the minimum distance path IS the zero-drift path (straight line across).

Drift formula for general angle theta (upstream from perpendicular):

$v_{\text{across}} = v_b \cos(\theta)$; $v_{\text{drift}} = v_r - v_b \sin(\theta)$

$t = d/(v_b \cos(\theta))$

$\text{Drift} = (v_r - v_b \sin(\theta)) * d / (v_b \cos(\theta)) = d(v_r/v_b - \sin(\theta))/\cos(\theta)$

Minimizing drift: $d(\text{drift})/d(\theta) = 0 \Rightarrow$ gives $\sin(\theta) = v_b/v_r$ (for $v_b < v_r$) or $\theta = \arcsin(v_r/v_b)$ for zero drift.

SECTION 3: RAIN PROBLEMS

Setup: Rain falls with velocity $v_{\text{rain_ground}}$. Person moves with velocity v_{man} .
Rain velocity as seen by person: $v_{\text{rain_man}} = v_{\text{rain_ground}} - v_{\text{man}}$.

Direction of umbrella: The person experiences rain coming from the direction of $v_{\text{rain_man}}$.

Umbrella must be tilted in the direction of $v_{\text{rain_man}}$ (toward the apparent rain source).

OR equivalently: umbrella tilted in direction of v_{man} (opposite to person's motion direction)... NO:

If v_{man} is to the east, $v_{\text{rain_man}}$ has a westward component (rain appears to come from the east/front).

So umbrella tilts forward (eastward, in direction of motion).

Vertical rain, horizontal man:

$v_{\text{rain_ground}} = (0, -v_r)$ [downward]. $v_{\text{man}} = (v_m, 0)$ [horizontal].

$v_{\text{rain_man}} = (0, -v_r) - (v_m, 0) = (-v_m, -v_r)$.

Rain appears to come from the east-and-upward direction. Umbrella tilts eastward.

Angle from vertical: $\tan(\alpha) = v_m/v_r$. So $\alpha = \arctan(v_m/v_r)$.

Inclined rain:

$v_{\text{rain_ground}} = (v_{rx}, -v_{ry})$ [some horizontal component + downward].

$v_{\text{man}} = (v_m, 0)$.

$v_{\text{rain_man}} = (v_{rx} - v_m, -v_{ry})$.

Umbrella angle from vertical: $\tan(\beta) = (v_{rx} - v_m)/v_{ry}$.

When is rain striking equally front and back? When $|v_{\text{rain_man}}|$ makes 45 deg from vertical.

This gives $\tan(45) = 1 = (v_m - v_{rx})/v_{ry} \Rightarrow v_m = v_{ry} + v_{rx}$.

SECTION 4: INERTIAL AND NON-INERTIAL FRAMES

Inertial Frame: A reference frame that is not accelerating.
Newton's laws hold in their standard form in inertial frames.
All frames moving at constant velocity relative to a known inertial frame are also inertial.
The ground frame (approximately) is inertial for most JEE problems.

Non-Inertial Frame: A reference frame that is accelerating.
To use Newton's laws in a non-inertial frame, add a pseudo force on EVERY body:
 $F_{\text{pseudo}} = -m * a_{\text{frame}}$ (on each body of mass m)
where a_{frame} = acceleration of the reference frame in an inertial frame.

Accelerating lift (elevator):

Lift accelerates upward at a : spring balance reads $W' = m*(g+a)$. Heavier.
Lift accelerates downward at a : spring balance reads $W' = m*(g-a)$. Lighter.
Lift in free fall ($a=g$): $W' = m*(g-g) = 0$. Weightlessness (apparent zero gravity).
Lift decelerating upward (a downward): $W' = m*(g-a)$. Same as moving down.

Car turning in a circle:

Centripetal acceleration: $a_{\text{frame}} = v^2/R$ (inward).
Pseudo force (centrifugal) = $m*v^2/R$ (outward). Passenger feels pushed outward.

Galilean transformation:

If frame S' moves at constant v relative to frame S :
 $x' = x - v*t$; $t' = t$ (Galilean, non-relativistic).
 $v'_{\text{object}} = v_{\text{object}} - v$ (velocity transformation).
 $a'_{\text{object}} = a_{\text{object}}$ (same acceleration in all inertial frames).
=> Newton's laws have the same form in all inertial frames. (Galilean invariance)

SECTION 5: MINIMUM DISTANCE AND COLLISION PROBLEMS

Minimum distance between two moving particles:

1. Compute $v_{AB} = v_A - v_B$ (relative velocity, treating B as origin).
2. Compute $r_{AB}(0) = r_A(0) - r_B(0)$ (initial relative position vector).
3. Find angle α between $r_{AB}(0)$ and v_{AB} vectors.
4. $d_{min} = |r_{AB}| \sin(\alpha)$.
5. Time: $t_{min} = |r_{AB}| \cos(\alpha) / |v_{AB}|$ (if $\cos(\alpha) > 0$, they approach).

Coefficient of restitution in oblique collisions:

For two bodies colliding, along the line of impact:

$$e = (v_{2f} - v_{1f}) / (v_{1i} - v_{2i}) \text{ [velocity of separation / velocity of approach along impact line]}$$

$e = 1$: perfectly elastic. $e = 0$: perfectly inelastic. $0 < e < 1$: partially elastic.

For head-on collision (1D) with e:

$$v_{1f} = ((m_1 - e m_2) v_{1i} + (1 + e) m_2 v_{2i}) / (m_1 + m_2)$$

$$v_{2f} = ((m_2 - e m_1) v_{2i} + (1 + e) m_1 v_{1i}) / (m_1 + m_2)$$

Relative velocity of approach and separation:

$$v_{approach} = v_{1i} - v_{2i} \text{ (before collision, in 1D).}$$

$$v_{separation} = v_{2f} - v_{1f} \text{ (after collision).}$$

$$\text{Restitution: } v_{separation} = e \cdot v_{approach}.$$

WORKED EXAMPLES — JEE ADVANCED LEVEL

Example 1: River Crossing — Minimum Time

River width $d=100$ m, $v_r=3$ m/s, $v_b=4$ m/s. Find min time and drift.

Aim boat directly across (perpendicular to banks).

$$t_{\min} = d/v_b = 100/4 = 25 \text{ s.}$$

$$\text{Drift} = v_r t = 3 \times 25 = 75 \text{ m (downstream).}$$

$$\text{Ground speed} = \sqrt{4^2 + 3^2} = 5 \text{ m/s at angle } \arctan(3/4) = 36.9^\circ \text{ from perpendicular.}$$

Answer: $t=25$ s, drift=75 m, boat travels at 5 m/s over ground.

Example 2: River — Zero Drift (Upstream Aim)

Same river: $d=100$ m, $v_r=3$ m/s, $v_b=4$ m/s. Find angle and time for zero drift.

$$\sin(\theta) = v_r/v_b = 3/4. \theta = 48.59^\circ \text{ upstream from perpendicular.}$$

$$\text{Effective across-river speed} = \sqrt{v_b^2 - v_r^2} = \sqrt{16-9} = \sqrt{7} = 2.646 \text{ m/s.}$$

$$t = d/\sqrt{7} = 100/2.646 = 37.8 \text{ s.}$$

Boat aims at 48.59 deg upstream from direct crossing direction. Takes 37.8 s with no drift.

Example 3: Rain and Umbrella Direction

Man walks east at 3 m/s. Rain falls vertically at 4 m/s. Umbrella direction?

$$v_{\text{rain_ground}} = (0, -4) \text{ m/s (downward). } v_{\text{man}} = (3, 0) \text{ m/s (east).}$$

$$v_{\text{rain_man}} = (0, -4) - (3, 0) = (-3, -4) \text{ m/s.}$$

$$\text{Magnitude} = \sqrt{9+16} = 5 \text{ m/s. Angle from vertical: } \tan(\alpha) = 3/4 = 0.75. \alpha = 36.9^\circ \text{ deg.}$$

Rain appears to come from the east. Umbrella tilts 36.9 deg toward east (forward, in direction of motion).

Answer: Umbrella tilted 36.9 deg forward (toward east/direction of motion).

Example 4: Minimum Distance Between Two Cars

Car A: 60 km/h north starting at origin. Car B: 80 km/h east, initially 500 m west of A.

Find minimum distance between them.

$$v_A = (0, 60) \text{ km/h, } v_B = (80, 0) \text{ km/h.}$$

$$v_{AB} = v_A - v_B = (-80, 60) \text{ km/h. } |v_{AB}| = 100 \text{ km/h.}$$

$$r_{AB}(0) = r_A - r_B = (0, 0) - (-0.5, 0) = (0.5, 0) \text{ km (A is 0.5 km east of B).}$$

$$\cos(\alpha) = r_{AB} \cdot v_{AB} / (|r_{AB}| |v_{AB}|) = (0.5 \times (-80) + 0 \times 60) / (0.5 \times 100) = -40/50 = -0.8.$$

$$\text{Wait: } r_{AB} = (0.5, 0), v_{AB} = (-80, 60). \text{ Dot} = 0.5 \times (-80) + 0 \times 60 = -40. \cos = -40/50 = -0.8.$$

$$\sin(\alpha) = \sqrt{1-0.64} = 0.6. d_{\min} = 0.5 \times 0.6 = 0.3 \text{ km} = 300 \text{ m.}$$

They are approaching ($\cos(\alpha) < 0$? No, that means diverging). Since $\cos < 0$: particles moving apart! Min dist is initial dist = 500 m.

Recheck: $r_{AB} \cdot v_{AB} < 0$ means A is getting farther from B. Min dist = 500 m (initial).

Example 5: Accelerating Lift

Person mass 60 kg in lift accelerating upward at 2 m/s^2 . Then decelerating at 2 m/s^2 (going up).

Find apparent weight in both cases. $g=10 \text{ m/s}^2$.

Case 1 (accelerating up, $a=+2$ upward): $W' = m(g+a) = 60(10+2) = 720 \text{ N}$ (reads 72 kg).

Case 2 (decelerating going up = acceleration downward at 2): $W' = m(g-a) = 60(10-2) = 480 \text{ N}$.

Case 3 (free fall): $W' = m(g-g) = 0 \text{ N}$. Weightlessness.

Answer: 720 N (up accel), 480 N (decelerate while going up), 0 N (free fall).

Example 6: Projectile in Moving Train

Ball thrown vertically upward at $u=20 \text{ m/s}$ in train moving at 30 m/s east.

Find: (a) trajectory in ground frame. (b) max height.

(a) In ground frame: $x = 30t$, $y = 20t - 5t^2$ (vertical).

Eliminating t : $t = x/30$. $y = 20(x/30) - 5(x/30)^2 = 2x/3 - x^2/180$.

This is a parabola opening downward. In train frame: ball goes straight up and down.

(b) Max height: $v_y = 20 - 10t = 0 \Rightarrow t=2 \text{ s}$. $h = 20 \cdot 2 - 5 \cdot 4 = 40 - 20 = 20 \text{ m}$.

Answer: Parabola $y = 2x/3 - x^2/180$ in ground frame. Max height = 20 m.

PRACTICE QUESTIONS — JEE ADVANCED LEVEL

Q1. River $d=80$ m, $v_r=3$ m/s, $v_b=5$ m/s. Minimum crossing time:

(A) 16 s (B) 20 s (C) 25 s (D) 12 s

Q2. For zero drift river crossing, boat should head:

(A) Downstream (B) Directly across (C) Upstream at angle $\arcsin(v_r/v_b)$ (D) Upstream at 45 deg

Q3. Rain falls at 30 deg to vertical. Man moving in same horizontal direction as rain horizontal component sees rain falling:

(A) More steeply (B) Less steeply (C) Vertically (D) Horizontally

Q4. Lift in free fall. Body of mass m . Spring balance reads:

(A) mg (B) $2mg$ (C) 0 (D) $mg/2$

Q5. Two trains approach each other: 60 km/h and 90 km/h on parallel tracks. Relative speed:

(A) 30 km/h (B) 90 km/h (C) 150 km/h (D) 75 km/h

Q6. Car A moves 40 m/s north, car B moves 30 m/s east, start same point. Separation after 1 s:

(A) 50 m (B) 70 m (C) 40 m (D) 30 m

Q7. River $v_r=4$ m/s, $v_b=3$ m/s. Can boat cross with zero drift?

(A) Yes, at angle $\arcsin(3/4)$ (B) No, $v_b < v_r$ (C) Yes, directly across (D) Only if river is calm

Q8. Man walks at v east, rain falls vertically at u . Umbrella angle from vertical:

(A) $\arctan(u/v)$ (B) $\arctan(v/u)$ (C) $\arcsin(v/u)$ (D) 45 deg

Q9. Ball thrown vertically in moving train. In ground frame the trajectory is:

(A) Vertical line (B) Circle (C) Parabola (D) Ellipse

Q10. Coefficient of restitution $e=0.8$. Ball hits wall at 10 m/s. Rebounds at:

(A) 8 m/s (B) 10 m/s (C) 12 m/s (D) 6.4 m/s

ANSWER KEY AND EXPLANATIONS

Q1 — (A) 16 s

$t_{\min} = d/v_b = 80/5 = 16$ s. Aim directly across.

Q2 — (C) Upstream at $\arcsin(v_r/v_b)$

For zero drift: resolve v_b upstream component = v_r . $\sin(\theta) = v_r/v_b$.

Q3 — (A) More steeply

Moving in the direction of horizontal rain component reduces apparent horizontal component $\Rightarrow$ rain appears more vertical (steeper).

Q4 — (C) 0

Free fall: $a = g$ downward. Apparent weight = $m(g-g) = 0$. Weightlessness.

Q5 — (C) 150 km/h

Opposite directions on parallel tracks: relative speed = $60+90 = 150$ km/h.

Q6 — (A) 50 m

After 1s: A at (0,40), B at (30,0). $d = \sqrt{30^2+40^2} = \sqrt{900+1600} = \sqrt{2500} = 50$ m.

Q7 — (B) No, $v_b < v_r$

Zero drift requires $v_b > v_r$. Here $3 < 4$, so boat is always carried downstream. Cannot cross straight.

Q8 — (B) $\arctan(v/u)$

$v_{\text{rain_man}} = (0,-u)-(v,0) = (-v,-u)$. Angle from vertical = $\arctan(v/u)$.

Q9 — (C) Parabola

Horizontal motion at train speed (constant) + vertical parabolic motion. Result: parabola.

Q10 — (A) 8 m/s

$e = v_{\text{rebound}}/v_{\text{approach}}$. $0.8 = v_{\text{rebound}}/10 \Rightarrow v_{\text{rebound}} = 8$ m/s.